Strand 4: Handling Data

Sub-strand 1: Data and Statistics

Lesson 1: Data Types, Frequency Tables, Stem-and-Leaf, and Descriptive Statistics

1. Curriculum Standard and Overview

- Strand: 4 — Handling Data
- Sub-strand: 1 — Data and Statistics
- Content Standard: B9.4.1.1 / B9.4.1.2
- Indicators:
- B9.4.1.1.1 — Select and justify a method to collect data (quantitative vs qualitative).
- B9.4.1.1.2 — Organise data (grouped/ungrouped) and present it in frequency tables, stem-and-leaf plots, pie charts.
- B9.4.1.2.2 — Analyse data using descriptive statistics (mean, median, mode).
- Mock-paper links: P2 Q1(c) (median with unknown), P2 Q1(d) (stem-and-leaf, mode, median), P2 Q2(b) (mean height), P2 Q2(d) (frequency table), P2 Q4(a) (pie chart), P1 Q7 (mean with unknown), P1 Q28 (mean of remaining nine), P1 Q34 (numerical data).

2. Preamble — What Is Data?

Data is information collected for a purpose — test marks, heights, favourite colours.

2.1 Two big families

Type	Meaning	Examples
Quantitative (numerical)	counts or measurements	height in metres, number of pens, marks
Qualitative (categorical)	labels / words	eye colour, type of car, favourite food

Mock P1 Q34: Which is an example of numerical data?

A. Eye color B. Height in metres C. Type of car D. Favourite food

Answer: B — Height in metres.

3. Organising Data

3.1 Frequency table

Count how often each value appears.

Mock P2 Q2(d): Marbles (x) vs number of students (f).

x	f	f(x)
1	4	4
2	5	10
3	?	42
4	9	36
5	?	30

6	2	12

Complete: f(3) = 42 ÷ 3 = 14; f(5) = 30 ÷ 5 = 6.

- Total students Sigma f = 4 + 5 + 14 + 9 + 6 + 2 = 40.
- Total marbles Sigma fx = 4 + 10 + 42 + 36 + 30 + 12 = 134.
- Highest number of marbles brought by a student: 6.

3.2 Stem-and-leaf plot

Split each number into a stem (leading digits) and a leaf (last digit). Leaves must be written in ascending order.

Stem	Leaf
3	8
4	0 2 5 8
5	4 4 6 7 7 7 8 8
6	1 3 3 4 6
7	2 3

Key: $5 \mid 4 = 54$

Mock P2 Q1(d): 20 marks: 68, 42, 63, 57, 66, 54, 72, 58, 57, 73, 61, 54, 63, 45, 38, 40, 57, 58, 56, 48

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Stem | Leaf
3 | 8
4 | 0 2 5 8
5 | 4 4 6 7 7 7 8 8
6 | 1 3 3 6 8
7 | 2 3
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- Mode = most frequent = 57 (appears 3 times).
- Median = average of 10th and 11th values.

Ordered 1–20: the 10th is 57, the 11th is 58. Median = 57 + 58/2 = 57.5.

4. Descriptive Statistics

Measure	How to find it	Notes
Mean	sum of values ÷ number of values	pulled by outliers
Median	middle value (or mean of two middles)	resistant to outliers
Mode	most frequent	may be none or several
Range	max – min	simple spread

4.1 Mean with an unknown (Mock P1 Q7)

Mean of 5, 8, 4, 7, 2x is 6.

$$\frac{5+8+4+7+2x}{} = 6 \Rightarrow 24 + 2x = 30 \Rightarrow 2x = 6 \Rightarrow x = 3$$

Answer: C — 3.

4.2 Mean of a part of the data (Mock P1 Q28)

Mean of ten numbers is 58; one number is 40. Mean of the other nine?

Total = 580. Remaining sum = 580 - 40 = 540. Mean = 540 ÷ 9 = 60.

Answer: D — 60.

4.3 Mean with algebraic values (Mock P2 Q1c)

Data: 2, 3, (4m-3), (3m+1), 11, 13. Median in ascending order is 6. Find m.

Six values → median is the mean of the 3rd and 4th when ordered. For the median to be 6, the two middle values must average to 6.

Try m = 1: values 2, 3, 1, 4, 11, 13. Ordered: 1, 2, 3, 4, 11, 13. Median = 3+4/2 = 3.5 — wrong, so try the next value. Try m = 2: 2, 3, 5, 7, 11, 13. Ordered: 2, 3, 5, 7, 11, 13. Median = 5+7/2 = 6 — correct, so the answer is m = 2.

Answer: m = 2.

4.4 New mean when data changes (Mock P2 Q2b)

Mean height of 28 students is 162 cm. A new girl of 149 cm joins.

Total = 28 × 162 = 4536. New total = 4536 + 149 = 4685. New count = 29. New mean = 4685 ÷ 29 = 161.55ldots ≈ 161.6 cm (to 1 d.p.).

Answer: 161.6 cm (2 d.p.: 161.55 cm).

5. Pie Charts

Each sector angle = category value/(total) × 360^circ.

Mock P2 Q4(a): Pupils: One 35, Two 45, Three 24, Four 33, Five 23, Six 20. Total = 180.

- Fraction in Primary Two = 45/180 = ¼.
- Percentage of Primary Three, Four and Five together = 24+33+23/180 × 100 = 80/180 × 100 = 44.44%.

6. Twenty Fully Worked Examples

Example 1 — Mock P1 Q34

Height in metres is numerical. B.

Example 2 — Mock P1 Q7

x = 3. C.

Example 3 — Mock P1 Q28

Mean of other nine = 60. D.

Example 4 — Mock P2 Q1(c)

m = 2.

Example 5 — Mock P2 Q2(b)

New mean = 161.6 cm (1 d.p.).

Example 6 — Mock P2 Q2(d)
Complete table: f(3) = 14, f(5) = 6; Sigma f = 40; Sigma fx = 134; highest = 6.

Example 7 — Mock P2 Q1(d)
Stem-and-leaf; mode 57; median 57.5.

Example 8 — Mock P2 Q4(a)
Two = ¼ of pupils; Three+Four+Five = 44.44%.

Example 9 — Simple mean
Mean of 3, 5, 7, 9 = 6.

Example 10 — Simple median
Median of 3, 5, 7, 9, 11 = 7.

Example 11 — Simple mode
Mode of 2, 3, 3, 5, 5, 5, 7 = 5.

Example 12 — Range
Range of 12, 18, 20, 25 = 13.

Example 13 — Mean of larger set
Mean of 2, 4, 6, 8, 10, 12 = 7.

Example 14 — Median of even count
Median of 2, 4, 6, 8 = 5.

Example 15 — Mean with unknown
Mean of x, 10, 12 is 11. Sum = 33; x = 11.

Example 16 — Pie chart angle
Category 30 out of 120: angle = 30/120 × 360 = 90^circ.

Example 17 — Frequency table total
f = 2, 3, 5; x = 1, 2, 3: Sigma fx = 2 + 6 + 15 = 23.

Example 18 — Stem-and-leaf key
4 | 3 means 43. Always state the key.

Example 19 — Qualitative data
"Type of car" is qualitative (categorical).

Example 20 — Combined
Marks 50, 60, 70, 80. Mean = 65; median = 65; range = 30.

7. Comprehensive Exercise Sets (40 Questions)

All 8 authentic mock data items are included. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Data types (includes mock)

1. [Mock 2026 P1 Q34] Which of the following is an example of numerical data?
   A. Eye color B. Height in metres C. Type of car D. Favourite food
2. Which is qualitative: "number of pens" or "type of pen"?
3. Give one example of discrete quantitative data.
4. Give one example of continuous quantitative data.
5. Spiral review: Find 12x + 9q when x = 11, q = 3.

Exercise 2 — Mean (includes mock)

1. [Mock 2026 P1 Q7] The mean of 5, 8, 4, 7 and 2x is 6. Find x.
   A. 6 B. 4 C. 3 D. 2
2. [Mock 2026 P1 Q28] The mean of ten numbers is 58. One of the numbers is 40. What is the mean of the other nine?
   A. 54 B. 56 C. 58 D. 60
3. [Mock 2026 P2 Q2(b)] The mean height of 28 students is 162 cm. A new girl of 149 cm joins. Find the new mean.
4. Find the mean of 3, 5, 7, 9.
5. Spiral review: 13x - 12 = 5x + 60. Find x.

Exercise 3 — Median & mode (includes mock)

1. [Mock 2026 P2 Q1(c)] Data: 2, 3, (4m-3), (3m+1), 11, 13. The median in ascending order is 6. Find m.
2. Find the median of 3, 5, 7, 9, 11.
3. Find the median of 2, 4, 6, 8.
4. Find the mode of 2, 3, 3, 5, 5, 5, 7.
5. Spiral review: Area of a semicircle with r = 7 (π = 22/7).

Exercise 4 — Stem-and-leaf (includes mock)

1. [Mock 2026 P2 Q1(d)] The marks of 20 students are: 68, 42, 63, 57, 66, 54, 72, 58, 57, 73, 61, 54, 63, 45, 38, 40, 57, 58, 56, 48.

(i) Draw a stem-and-leaf plot in ascending order. (ii) What is the mode? (iii) Determine the median.

2. Using the same data, find the range.
3. Using the same data, find the mean.
4. Spiral review: If the area of a circle is 64π, find the circumference.
5. Spiral review: A hall 20 m long is drawn as 10 cm. What is the scale?

Exercise 5 — Frequency tables (includes mock)

1. [Mock 2026 P2 Q2(d)] Complete the table: x = 1,2,3,4,5,6; f = 4,5,?,9,?,2; f(x) has 42 at x=3, 30 at x=5, 12 at x=6.

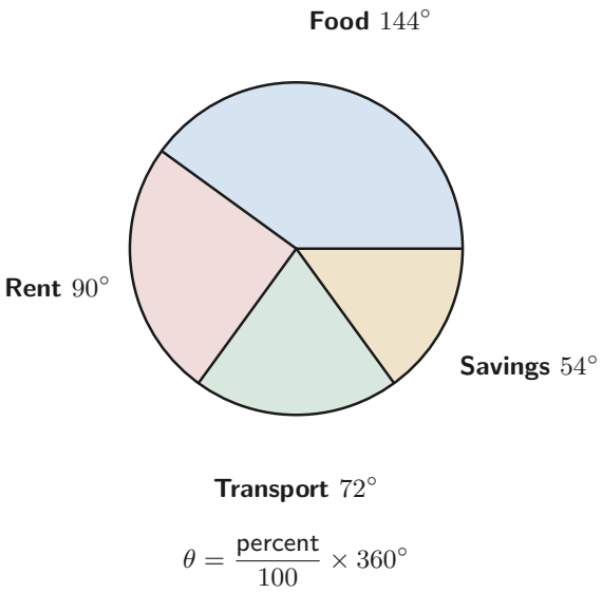
(i) Complete the table. (ii) How many students were in the class? (iii) How many marbles were brought altogether? (iv) What is the highest number of marbles brought by a student?

2. If f = 2, 3, 5 and x = 1, 2, 3, find Sigma fx.
3. Spiral review: 2x + 1 > 9. Solve.
4. Spiral review: Classify −3 under ℕ, 𝕎, ℤ, ℚ.
5. Spiral review: begin{pmatrix} 3 \\ 5 end{pmatrix} + begin{pmatrix} −1 \\ 2 end{pmatrix} = ?

Exercise 6 — Pie charts (includes mock)

1. [Mock 2026 P2 Q4(a)] Pupils: One 35, Two 45, Three 24, Four 33, Five 23, Six 20.

(i) Draw a pie chart for the data.



$$\theta = \frac{\text{percent}}{100} \times 360^\circ$$

(ii) What fraction of the pupils are in primary two? (iii) Calculate the percentage of pupils in primary three, four and five altogether, correct to 2 decimal places.

2. A category has 30 out of 120 items. Find its pie chart angle.
3. Spiral review: (−3)^(x-2) = −243. Find x.
4. Spiral review: A van travels 154 km in 1 ¾ h. Speed?
5. Spiral review: n/100 = 10.5. Find n.

Exercise 7 — Range & spread

1. Find the range of 12, 18, 20, 25.
2. Find the mean, median and range of 50, 60, 70, 80.
3. Find the mean of 2, 4, 6, 8, 10, 12.
4. Spiral review: Solve 4/(3x) = 7 - 3/x.
5. Spiral review: Three consecutive odd integers sum to 33.

Exercise 8 — Mixed statistics challenge

1. Mean of x, 10, 12 is 11. Find x.
2. Median of 3, 7, x, 15 is 9. Find x.
3. A set has mean 20 and 5 values. Four of them are 15, 18, 22, 25. Find the fifth.
4. Spiral review: In a market of 36 fishmongers, 28 sell mackerel, 18 tilapia, 13 both. How many sell neither?
5. Spiral review: Find the gradient through (1, 2) and (4, −4).

8. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. B — Height in metres.
2. Type of pen (qualitative).
3. e.g. number of pupils in a class.
4. e.g. height in metres.
5. 159.

Solutions to Exercise 2

1. $24 + 2x = 30 \rightarrow x = 3 \rightarrow$ C.
2. $(580 - 40) \div 9 = 60 \rightarrow$ D.
3. $(4536 + 149) \div 29 = 161.6$ cm (1 d.p.).
4. 6.
5. $x = 9$.

Solutions to Exercise 3

1. $m = 2$ (check: ordered 2,3,5,7,11,13, median 6).
2. 7.
3. 5.
4. 5.
5. 77 cm^2.

Solutions to Exercise 4

1. (i) stems 3–7 as above; (ii) mode 57; (iii) median 57.5.
2. Range = 73 - 38 = 35.
3. Sum = 1097; mean = $1097 \div 20 = 54.85$.
4. 16π.
5. 1 : 200.

Solutions to Exercise 5

1. (i) $f(3) = 14$, $f(5) = 6$; (ii) Sigma f = 40; (iii) Sigma fx = 134; (iv) 6.
2. Sigma fx = 2 + 6 + 15 = 23.
3. $x > 4$.
4. $\mathbb{Z}$, $\mathbb{Q}$.
5. begin{pmatrix} 2 \\ 7 end{pmatrix}.

Solutions to Exercise 6

1. (ii) $45/180 = 1/4$; (iii) $80/180 \times 100 = 44.44\%$.
2. 90^circ.
3. $x = 7$.
4. 88 km/h.
5. $n = 1050$.

Solutions to Exercise 7

1. 13.
2. Mean = 65; median = 65; range = 30.
3. 7.
4. $x = 13/21$.
5. 9, 11, 13.

Solutions to Exercise 8

1. $x = 11$.
2. $(7 + x) / 2 = 9 \rightarrow x = 11$.
3. $100 - 80 = 20$.
4. 3.
5. $m = -2$.

9. Authentic Mock Items in This Lesson

Ref	Question	Answer
P1 Q34	Numerical data example	B — Height in metres
P1 Q7	Mean of 5, 8, 4, 7, 2x is 6	C — x = 3
P1 Q28	Mean of ten is 58, one is 40	D — 60
P2 Q1(c)	Median 6 with unknowns	m = 2
P2 Q1(d)	Stem-and-leaf of 20 marks	mode 57; median 57.5
P2 Q2(b)	New mean after girl joins	161.6 cm
P2 Q2(d)	Marbles frequency table	Sigma f = 40, Sigma fx = 134
P2 Q4(a)	Pupils pie chart	Two = 1/4; Three+Four+Five = 44.44%

10. Lesson Summary

Skill	Rule
Mean	sum ÷ count
Median	middle (mean of two middles if even)
Mode	most frequent
Range	max − min
Pie chart angle	(value ÷ total) × 360°
Stem-and-leaf	stems + ordered leaves; state the key

Lesson prepared under `Grade9-Maths/Rules.md` — plain English, British notation, full workings required.